Project Proposal

NYC Ride-Hailing, Weather & EV Charging Access

Course	DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore
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Phase	1 - Business Analysis & Dimensional Design

1. Project Overview

We will build a data warehouse and BI pipeline using only real public data about ride-hailing in New York City. It brings together trips, hourly weather and EV charging stations, so planners can analyse them in one place. The business problem, scope, requirements and stakeholders are set out in the Business Requirements Document.

2. Project Objectives

- Design a star schema with three fact tables, using the grain-first approach from Lab 05.
- Build an ETL pipeline from raw data to cleaned data, the warehouse and data marts.
- Keep history with SCD Type 1 and Type 2 where it matters.
- Answer the business questions with OLAP SQL (ROLLUP, CUBE, window functions) and a Power BI dashboard.

3. Pipeline Architecture

The pipeline follows the teacher's reference diagram, in nine stages.

#	Stage	What happens
1	Sources	PostgreSQL trips, Open-Meteo API, AFDC station API (JSON), taxi zone CSV.
2	Bronze	Raw data loaded as it is, with batch ID and load time.
3	Data quality checks	NULL, data type, duplicate and referential checks. Bad rows go to a reject table with a reason.
4	Silver	Cleaned and standardised data, loaded incrementally. Each station is placed in a zone with a one-time spatial join (Python / GeoPandas).
5	SCD processing	Detect changes, then apply Type 1 or Type 2.
6	Data Warehouse	Dimensions and facts, with a late-arriving data plan (unknown member rows).
7	Gold	Data marts for trips, zone demand and charging coverage.
8	Analytics and BI	OLAP SQL and a Power BI dashboard with spatial maps.
9	Audit and lineage	Run logs, row counts, and source vs warehouse reconciliation.

4. Known Challenges and Our Plan

Challenge	Our Plan
Trip data is very large (millions of rows).	Work on a 500k-row sample to keep PostgreSQL fast.
Zone matching can cause Cartesian joins.	Join through unique surrogate keys on Dim_Zone.
Type 2 history needs more than one station download.	Load an older AFDC station snapshot first and the latest data after it, so changes create new row versions in Dim_Station.

5. Tools

PostgreSQL, pgAdmin, Python/Pandas (with GeoPandas for the zone join), Power BI Desktop, Draw.io / dbdiagram.io, GitHub.

6. Timeline

Phase	Weeks	Work	Weight
1: Design	1-3	Proposal, requirements, grain, schema, data dictionary	30%
2: Implementation	4-6	ETL, SCD, dimensions and facts	40%
3: Analytics & BI	7-8	OLAP SQL, Power BI dashboard	30%

7. Expected Outcome

A working warehouse and dashboard showing three clear findings: how rain changes demand, wait time and driver pay; which zones and hours are busiest; and where busy zones have too few chargers.